

Model 1 - Baseline Results

Overall model:

- $R^2 = 0.089$ - the model explains 8.9% of variance in LogReviewCount. Modest but expected for marketing data with many unobserved factors
- F-statistic = 72.86, $p < 0.001$ - the model as a whole is highly significant

SocialImpact ($\beta_1 = 0.220$, $p = 0.450$) - NOT significant

- The sign is **positive** - cause marketing is directionally associated with MORE demand
- But the wide confidence interval $[-0.35, 0.79]$ tells the real story - we simply cannot estimate this effect precisely with only 35 social impact listings
- This is expected and not a failure - it will be flagged as a limitation in the writeup

LogPrice ($\beta = -0.255$, $p < 0.001$) - Significant

- Higher priced listings get significantly fewer reviews - makes intuitive sense

average_rating ($\beta = 0.075$, $p < 0.001$) - Significant

- Higher rated listings get more reviews - also intuitive

Category effects:

- Shoes (-1.17) and Art & Collectibles (-1.15) get significantly fewer reviews than Weddings
- Toys & Games (-0.63) and Pet Supplies (-0.39) also significantly negative vs Weddings
- No category significantly outperforms Weddings

Model 2 - Sentiment Model Results

Overall model:

- R^2 jumped from 0.089 $\rightarrow$ 0.106, a gain of +0.016 - SentimentScore meaningfully improves the model
- F-statistic = 81.83, $p < 0.001$ - model highly significant overall

SentimentScore ($\beta_2 = 1.055$, $p < 0.001$) - Highly Significant

- The strongest single predictor in the model
- A one-unit increase in sentiment score (e.g., from 0 to 1) is associated with 1.055 higher LogReviewCount
- Since most listings cluster between 0.8–1.0 on sentiment, even small differences in tone translate to meaningful differences in reviews
- Positive language independently predicts more demand

SocialImpact ($\beta_1 = 0.176$, $p = 0.542$) - Still Not Significant

- Coefficient actually dropped slightly from 0.220 $\rightarrow$ 0.176 after controlling for sentiment
- This means some of the raw association between social impact and reviews was being driven by the positive language in those descriptions - not the cause marketing itself
- Still positive directionally - cause marketing is not hurting demand

Model 3 Results - Social Impact Moderation Model

$R^2 = 0.1057$ - identical to Model 2, the interaction term added zero explanatory power.

β_3 (Interaction) = **-0.214**, $p = 0.913$ - **Not Significant** The wide confidence interval [-4.04, 3.62] tells the whole story - we simply cannot estimate the moderation effect with any precision. With only 35 social impact listings, the interaction term is massively underpowered.

Moderation Plot - the two lines are nearly parallel, confirming no meaningful interaction.

Model 4 Results - HandMade Model

Overall model:

- $R^2 = 0.109$ - improved from Model 2's 0.106, a gain of +0.004
- F-statistic = 79.73, $p < 0.001$ - model highly significant
- All VIF < 3 - no multicollinearity concerns

Handmade ($\beta_3 = -0.235$, $p < 0.001$) - Highly Significant This is the headline finding. Handmade listings get significantly **fewer** reviews than non-handmade listings, even after controlling for price, rating, category and sentiment. A coefficient of -0.235 on LogReviewCount means handmade listings have approximately 21% fewer reviews on average than comparable non-handmade listings.

It can be due to obvious reasons according to me such as because they are handmade - typically produced in small quantities, higher pricing due to manual effort, less likely to be repeat purchases, some handmade orders are made-to-order targeting specific users, niche market scope.

SentimentScore ($\beta_2 = 1.112$, $p < 0.001$) - Consistent and stronger The sentiment effect actually increased slightly from 1.055 in Model 2 to 1.112 here - adding Handmade clarified the sentiment effect rather than absorbing it.

SocialImpact ($\beta_1 = 0.154$, $p = 0.593$) - **Still not significant** Consistent with all previous models. The coefficient is still positive but underpowered.

Model 5 Results - HandMade Moderation Model

$R^2 = 0.1094$ - identical to Model 4, interaction term adds zero explanatory power (+0.0001).

Handmade ($\beta_3 = -0.231$, $p < 0.001$) - **Highly Significant** - The main effect holds perfectly after adding the interaction term.

Handmade $\times$ Sentiment ($\beta_5 = -0.158$, $p = 0.395$) - **Not Significant** The interaction does not reach significance. However unlike SocialImpact where we had only 35 listings making the test completely underpowered, here we have 3,671 handmade listings - this is a genuine null result, not a power issue.

The moderation plot confirms it - the two lines are nearly parallel, meaning positive language benefits handmade and non-handmade listings equally.

Interpretation for the Above Models only :

1. Does cause marketing work (β_1)? Directionally yes - positive across all three models - but statistically underpowered due to only 35 social impact listings (0.34% of dataset).

2. Does sentiment predict demand (β_2)? Yes, strongly and consistently - $\beta = 1.055$, $p < 0.001$ across Models 2 and 3. This is the most robust finding.

3. Does sentiment moderate the social impact effect (β_3)? Cannot be determined - the interaction is not significant and the confidence interval is extremely wide. More social impact listings would be needed to test this properly.

4. Does handmade status predict demand? (β_3) Yes - strongly and significantly negative ($\beta_3 = -0.231$ to -0.235 , $p < 0.001$ across Models 4 and 5). Handmade listings consistently get fewer reviews than non-handmade listings after controlling for price, rating, category and sentiment. This is a genuine, well-powered finding.

5. Does sentiment predict demand? (β_2) Yes - consistently positive and highly significant across all models where it appears ($\beta_2 = 1.055$ to 1.112 , $p < 0.001$). Most robust finding in the entire analysis.

6. Does sentiment moderate the handmade effect? (β_5) No - $\beta_5 = -0.158$, $p = 0.395$. Unlike the SocialImpact moderation test which was underpowered, this is a genuine null result with 3,671 handmade listings. Sentiment benefits all listings equally regardless of handmade status.

New Feature Extraction :

Lexical Diversity ($r = -0.195$) - Most surprising finding This is the strongest correlation of all five features and it's negative. Listings with more varied vocabulary actually get fewer reviews. This seems counterintuitive but makes sense on Etsy - short, simple, keyword-heavy listings (low lexical diversity) likely rank better in Etsy's search algorithm and get more traffic. Highly literary, varied descriptions may be harder to search for. This is analytically meaningful and worth testing in regression.

Trust Factor ($r = +0.125$) - Strong positive signal Second strongest correlation. Listings with free returns, money back guarantees, fast shipping language get more reviews. Makes intuitive sense - reduced purchase risk = more buyers = more reviews. Comparable in strength to SentimentScore ($r = 0.13$). This is likely to be significant in regression.

Personalization ($r = +0.078$) - Moderate positive Custom/made-to-order listings get more reviews. Also intuitive - personalized products tend to have more engaged buyers who are more motivated to leave feedback. Worth testing.

First Person Ratio ($r = +0.021$) - Weak Very weak correlation. Seller storytelling voice doesn't meaningfully predict demand on its own. May still work as a moderator but is unlikely to be significant as a main effect.

Readability Score ($r = -0.036$) - Weak negative Slightly more readable descriptions get fewer reviews - consistent with the lexical diversity finding. Simple, scannable product spec listings may outperform well-written prose on Etsy's platform.

Some notes from their distributions as well :

Readability Score has a problematic range - minimum of -138.5 is impossible on the Flesch scale (0-100). This happens with very short descriptions or descriptions with unusual formatting (bullet points, HTML entities, numbers). Will need winsorizing or clipping before regression.

First Person Ratio is heavily right-skewed - 4,200+ listings have near-zero ratio, meaning most Etsy sellers write clinically without personal voice.

Trust Factor at only 10.75% - surprisingly low. Most Etsy sellers don't explicitly mention returns or shipping guarantees in their descriptions, which makes the 1,119 that do potentially interesting.

Model 6 Results - Lexical Diversity Model

Overall model:

- $R^2 = 0.140$ - the best R^2 across all models so far, up from 0.109 in Model 4
- R^2 change = **+0.030** - largest single jump in explanatory power in the entire analysis
- F-statistic = 99.14, $p < 0.001$ - highly significant over

LexicalDiversity ($\beta_4 = -0.342$, $p < 0.001$) - **Highly Significant** This is a strong and clean finding. Every one standard deviation increase in lexical diversity is associated with a 0.342 decrease in LogReviewCount, holding everything else constant. The predicted demand bar chart makes this tangible - listings at low lexical diversity (-1 SD) are predicted to have LogReviewCount of 7.095 compared to 6.411 for high lexical diversity (+1 SD), a difference

of **0.684**. In plain terms - simple, repetitive, keyword-heavy descriptions significantly outperform richly written ones on Etsy.

SentimentScore ($\beta_2 = 0.535$, $p < 0.001$) - Significant but dropped This is critical - SentimentScore coefficient dropped from 1.055 in Model 2 to 0.535 here. This confirms what the correlation heatmap warned us about ($r = -0.380$ between lexical diversity and sentiment). Part of what we previously attributed to positive sentiment was actually being driven by low lexical diversity - simple positive language (low diversity) was doing double duty in earlier models. Now that both are in the model, each gets its proper share of the explanatory power.

Handmade ($\beta_3 = -0.283$, $p < 0.001$) - Significant and strengthened The handmade effect actually got stronger (-0.235 in Model 4 vs -0.283 here). Handmade listings tend to have slightly lower lexical diversity, so controlling for lexical diversity reveals the pure handmade penalty is larger than we previously thought.

SocialImpact ($\beta_1 = 0.045$, $p = 0.873$) - Not significant The SocialImpact coefficient dropped dramatically from 0.154 in Model 4 to 0.045 here. Social impact listings have lower lexical diversity (0.636 vs 0.691 from Task 4c) - meaning some of the previously observed positive direction of SocialImpact was being driven by their simpler, more repetitive cause marketing language rather than the cause marketing itself.

Model 7 Results - Lexical Diversity Moderation Model

$R^2 = 0.1397$ - identical to Model 6, interaction term adds zero explanatory power (+0.0001).

LexDiv $\times$ Sentiment ($\beta_5 = -0.583$, $p = 0.286$) - Not Significant Sentiment does not moderate the lexical diversity effect. The two lines on the moderation plot are nearly parallel - positive language benefits low and high lexical diversity listings equally. This is a genuine null result - with 10,406 listings we have more than enough power to detect an interaction if one existed.

LexicalDiversity ($\beta_4 = -2.957$, $p < 0.001$) - Highly Significant and Robust The main effect holds strongly after adding the interaction term. Every unit increase in lexical diversity (mean-centered) is associated with a 2.957 decrease in LogReviewCount. The effect is large, consistent, and clearly not driven by sentiment.

SentimentScore ($\beta_2 = 0.590$, $p < 0.001$) - Significant and Stable Virtually unchanged from Model 6 (0.535 $\rightarrow$ 0.590). Sentiment and lexical diversity work completely independently on demand.

Does lexical diversity predict demand? - Yes - strongly negative ($\beta_4 = -0.342$, $p < 0.001$)

Does sentiment moderate lexical diversity? - No - genuine null ($\beta_5 = -0.583$, $p = 0.286$)

Model 8 Results - Personalization Model

$R^2 = 0.112$ - modest improvement from Model 4 (0.109), gain of +0.003

Personalization ($\beta_4 = +0.201$, $p < 0.001$) - Significant Personalized listings get significantly more reviews than non-personalized listings after controlling for everything else. The predicted demand bar chart shows a clean difference - Personalization=1 listings are predicted at 6.857 vs 6.655 for Personalization=0, a gap of +0.201. Listings that explicitly advertise customization options attract more buyers and more reviews, independent of their handmade status, sentiment, price, rating and category.

SentimentScore ($\beta_2 = 1.055$, $p < 0.001$) - Stable and consistent Virtually identical to Model 2 (1.055 $\rightarrow$ 1.055). Adding personalization did not absorb any of the sentiment effect, confirming they work as completely independent demand drivers.

Handmade ($\beta_3 = -0.255$, $p < 0.001$) - Significant and slightly reduced Dropped slightly from -0.235 in Model 4 to -0.255... wait actually it became slightly more negative. This makes sense - handmade listings are more personalized (43.3% vs 31.5%), so controlling for personalization reveals the pure handmade penalty is slightly larger than before because we are no longer giving handmade credit for the personalization benefit it carries.

SocialImpact ($\beta_1 = 0.158$, $p = 0.581$) - Consistent with all previous models Still positive, still not significant. Unchanged from Model 4.

Model 9 Results - Personalization Moderation Model

$R^2 = 0.1121$ - identical to Model 8, interaction adds literally zero explanatory power (+0.0000).

Personalization $\times$ Sentiment ($\beta_5 = -0.000$, $p = 0.998$) - Not Significant This is about as clear a null result as you can get. A coefficient of -0.0004 with $p = 0.998$ means sentiment has absolutely no moderating effect on personalization. The moderation plot confirms this perfectly - the two lines are exactly parallel, maintaining a constant gap of ~ 0.20 across all sentiment levels. Personalization works equally well regardless of how positively the description is written.

Personalization ($\beta_4 = +0.201$, $p < 0.001$) - Robust. Main effect holds perfectly after adding the interaction term. Identical to Model 8. Personalized listings consistently get more demand independent of sentiment.

Everything else unchanged - SentimentScore, Handmade, SocialImpact, and controls are all identical to Model 8. Adding the interaction disturbed nothing.

Does personalization predict demand? Yes - positive ($\beta_4 = +0.201$, $p < 0.001$)

Does sentiment moderate personalization? No - absolute null ($\beta_5 = -0.000$, $p = 0.998$)

Model 10 Results - Trust Factor Model

$R^2 = 0.123$ - solid improvement from Model 4 (0.109), gain of **+0.013**

TrustFactor ($\beta_4 = +0.673$, $p < 0.001$) - Highly Significant This is the second strongest predictor in the entire analysis after Lexical Diversity. Listings with trust language get 0.673 higher LogReviewCount than listings without - the predicted bar chart shows Trust=1 at 7.331 vs Trust=0 at 6.658, a gap of **+0.673**. In practical terms this is enormous - corresponding to roughly double the raw reviews. Professional sellers who explicitly address buyer risk concerns in their descriptions massively outperform those who don't.

SocialImpact ($\beta_1 = -0.076$, $p = 0.789$) - FLIPPED NEGATIVE This is the most important finding in Model 10. SocialImpact coefficient crossed zero for the first time in the entire analysis - from +0.154 in Model 4 to **-0.076** here. This confirms exactly what we predicted from Task 4c - the previously observed positive SocialImpact direction was being partially driven by the trust language that cause marketing sellers disproportionately use (45.7% of SI=1 vs 10.6% of SI=0). Once we control for Trust Factor, SocialImpact loses even its directional positive signal.

SentimentScore ($\beta_2 = 1.011$, $p < 0.001$) - Stable Dropped slightly from 1.112 in Model 4 to 1.011 here, consistent with the moderate correlation between trust and sentiment ($r = +0.106$). Still highly significant and robust.

Handmade ($\beta_3 = -0.243$, $p < 0.001$) - Stable Slightly reduced from -0.235 in Model 4 to -0.243 - essentially unchanged. Handmade effect is independent of trust language.

The SocialImpact flip is a key finding - it reframes the entire interpretation of the social impact effect. What appeared to be a positive (though insignificant) cause marketing signal across Models 1–5 was partly a trust language signal in disguise. When we properly control for trust language, cause marketing has essentially zero - and now slightly negative - association with demand.

Model 11 Results - Trust Factor Moderation Model

$R^2 = 0.1227$ - identical to Model 10, interaction adds zero explanatory power (+0.0001).

TrustFactor $\times$ Sentiment ($\beta_5 = +0.687$, $p = 0.375$) - Not Significant The interaction does not reach significance despite having a positive coefficient. The wide confidence interval [-0.830, 2.203] tells the story - there is simply not enough variation in the trust $\times$ sentiment combination to estimate this precisely. The moderation plot shows the two lines are not parallel - Trust=1 has a steeper slope than Trust=0 suggesting a possible amplification effect - but it falls short of statistical significance.

TrustFactor ($\beta_4 = +0.627$, $p < 0.001$) - Robust Main effect holds strongly after adding the interaction. Slightly lower than Model 10 ($0.673 \rightarrow 0.627$) as the interaction absorbed a small portion of the trust variance - expected behavior.

SocialImpact ($\beta_1 = -0.086$, $p = 0.764$) - Stays negative Confirms the flip from Model 10. SocialImpact remains negative after adding the trust interaction, further solidifying that the previously positive SocialImpact direction was trust language in disguise.

SentimentScore ($\beta_2 = 1.078$, $p < 0.001$) - Slightly increased Rose from 1.011 in Model 10 to 1.078, consistent with the interaction term partially separating the sentiment and trust effects.

Does trust factor predict demand? - Yes - strongly positive ($\beta_4 = +0.673$, $p < 0.001$)

Does sentiment moderate trust factor? - No - not significant ($\beta_5 = +0.687$, $p = 0.375$)

Interpretation for the New Feature Extracted Models only :

1. Does lexical diversity predict demand? (β_4) Yes - strongly and significantly negative ($\beta_4 = -0.342$, $p < 0.001$ in Model 6). Listings with simpler, more repetitive keyword-heavy language get significantly more reviews than those with richly varied vocabulary, after controlling for sentiment, handmade status, price, rating and category.

2. Does sentiment moderate the lexical diversity effect? (β_5) No - $\beta_5 = -0.583$, $p = 0.286$. Unlike the SocialImpact moderation test which was underpowered, this is a genuine null result with 10,406 listings providing full statistical power. Positive language benefits low and high lexical diversity listings equally - sentiment and lexical diversity are completely independent demand drivers. The moderation plot confirms this with nearly parallel lines across all sentiment levels.

3. Does personalization predict demand? (β_4) Yes - significantly positive ($\beta_4 = +0.201$, $p < 0.001$ in Model 8). Listings that explicitly advertise customization options get more reviews than non-personalized listings after controlling for all other variables.

4. Does sentiment moderate the personalization effect? (β_5) No - $\beta_5 = -0.000$, $p = 0.998$. This is the clearest null result in the entire analysis. A coefficient of essentially zero with $p = 0.998$ means sentiment has absolutely no moderating influence on the personalization effect. The moderation plot shows perfectly parallel lines with a constant gap of exactly $+0.201$ across all sentiment levels.

5. Does trust factor predict demand? (β_4) Yes - strongly and significantly positive ($\beta_4 = +0.673$, $p < 0.001$ in Model 10). This is the second largest predictor in the individual model series after lexical diversity. Listings with explicit buyer reassurance language - free returns, money-back guarantees, fast shipping - get substantially more reviews, corresponding to roughly double the raw review count.

6. Does sentiment moderate the trust factor effect? (β_5) No - $\beta_5 = +0.687$, $p = 0.375$. The interaction does not reach significance. The moderation plot does show a slightly steeper slope for Trust=1 listings suggesting a possible amplification direction, but the wide confidence interval $[-0.830, 2.203]$ means this cannot be estimated with any precision. Trust Factor and Sentiment work as completely independent demand drivers - the benefit of trust language is the same regardless of how positively the description is written, and the benefit of positive language is the same regardless of whether trust signals are present.

Model 12 Results -Final Comprehensive Model

Overall model:

- $R^2 = 0.1481$ - the best fit across all 21 models, explaining 14.8% of variance in LogReviewCount
- R^2 gain of +0.039 over Model 4 - combining all predictors adds meaningful explanatory power
- F-statistic = 95.06, $p < 0.001$

What Predicts Etsy Listing Popularity

1. Trust Factor ($\beta = +0.542$, $p < 0.001$) - #1 Predictor The strongest independent predictor of demand. When everything else is held constant, listings with explicit buyer reassurance language get 0.542 higher LogReviewCount than those without. Professional sellers who address buyer risk concerns in their descriptions massively outperform those who don't. Notably the coefficient shrank from +0.673 in Model 10 to +0.542 here - some of the trust effect was previously absorbing variance that properly belongs to lexical diversity and sentiment.

2. Sentiment Score ($\beta = +0.493$, $p < 0.001$) - #2 Predictor Positive, enthusiastic language is the second strongest demand driver. The coefficient shrank dramatically from 1.055 in Model 2 to 0.493 here - confirming that in earlier models sentiment was absorbing the effects of lexical diversity and trust, which are correlated with it. The true independent sentiment effect is about half what earlier models suggested.

3. Lexical Diversity ($\beta = -0.312$, $p < 0.001$) - #3 Predictor Simple, repetitive keyword-heavy descriptions outperform richly written ones even after controlling for all other linguistic features simultaneously. This is a robust, clean finding.

4. Handmade ($\beta = -0.289$, $p < 0.001$) - #4 Predictor Handmade listings consistently get fewer reviews. The coefficient actually strengthened slightly from -0.235 in Model 4 to -0.289 here, confirming this is a genuine, independent demand penalty not explained by any linguistic characteristic.

5. LogPrice ($\beta = -0.267$, $p < 0.001$) - #5 Predictor Higher priced listings get fewer reviews - consistent across all models throughout the analysis.

6. Rating ($\beta = +0.050$, $p = 0.002$) - Significant but small Higher rated listings get slightly more reviews. The small coefficient reflects the ceiling effect we identified in EDA - 80.5% of listings are rated above 4.8.

Two Notable Casualties that I noticed -

Personalization ($\beta = +0.037$, $p = 0.298$) - Lost significance Personalization was significant in isolation ($\beta = +0.201$, $p < 0.001$ in Model 8) but loses significance when Trust Factor enters the model simultaneously. This is because personalization and trust factor are correlated ($r = +0.073$) and trust factor is a stronger signal. Once the model controls for trust language, the personalization effect largely disappears. The independent contribution of personalization to demand is near zero when all other factors are accounted for.

SocialImpact ($\beta = -0.130$, $p = 0.645$) - Confirmed null Stays negative and non-significant as in Models 10 and 11. The cause marketing effect is fully explained by the professional communication style of those sellers - trust language, simpler vocabulary, positive language - not the cause itself.

Rank	Predictor	β	Direction	Significance
1	Trust Factor	+0.542	More demand	***
2	Sentiment Score	+0.493	More demand	***
3	Lexical Diversity	-0.312	Less demand	***
4	Handmade	-0.289	Less demand	***
5	Price	-0.267	Less demand	***
6	Rating	+0.050	More demand	**
-	Personalization	+0.037	-	ns
-	Social Impact	-0.130	-	ns

Etsy listing popularity is driven by HOW sellers communicate rather than WHAT they are selling. The two biggest demand drivers are trust-building language and positive emotional tone - both are writing choices entirely within a seller's control. Conversely, simple repetitive language outperforms literary descriptions, suggesting Etsy's search-driven discovery model rewards keyword density over writing quality. Being handmade reduces demand - likely

reflecting niche targeting and limited production scale - while cause marketing shows no independent effect once communication style is properly controlled for.